

Request for Solutions Brief (RFSB) Revision 1

Human-Machine Integrated Formation (HMIF)

Robotic Autonomous System (RAS) Software Prototype and Integration

Background. The U.S. Army's Rapid Capabilities and Critical Technologies Office (RCCTO) would like to invite interested entities to provide technical inputs, solutions, or concept descriptions to the objectives and requirements outlined herein. The Army has designated a RCCTO and Army Futures Command (AFC) team to prototype integrated solutions for robotics to develop and manage a human-machine integrated formation (HMIF) architecture that will guide emerging robotic autonomy systems (RAS) capabilities. This team will prototype, test, and evaluate solutions using continuous Soldier evaluations (CSE) that will enable valuable warfighter feedback with the systems to enable lessons learned and drive system improvements. They will fully characterize and address any issues associated with the software, network, and command and control (C2) aspects of robotic systems, the modular mission payloads, and HMIF interaction and integration requirements. They will define priorities and consistent acquisition pathways for robotic systems; align operational use cases with the maturation roadmap of enabling technology and robotic behaviors; and promote rapid procurement of the latest industry RAS and enabler investments to fill critical technology gaps.

To that end, RCCTO is responding to a need for **HMIF Command and Control software, Ground Autonomy Software, Assured Control and Safety Software, Mission Command Interconnect Software, and Lead Software Integrator** that can coexist and interoperate with other existing tactical networks and HMIF platforms and payloads to provide the software that will drive time sensitive robotic command and control data, to include critical safety data, intermixed with the general transport of sensor or other payload data to relevant tactical consumers of that data. This software could be thought of as the robotic command and control extension to the tactical network – integrating the unique data requirements arising from robotic system command and control with the human formation that operates and consumes the information from those robots. RCCTO and other government subject matter experts (SMEs), including contractor support personnel, will review responses for software solutions and prototypes, including software integration, to better inform and shape the requirements for future work. This RFSB is open to both traditional and nontraditional large and small business concerns, as well as academic and nonprofit entities to submit their technical inputs, software development capabilities, solutions, and prototypes that may meet the Army's needs and objectives.

This request for solutions brief (RFSB) is an opportunity for industry partners to present capabilities, prototypes, concepts, and HMIF software solutions directly to the U.S. Army that fit within this mission and the stated objectives. The identification of team members/subcontractors

and the role that they play in delivering the software solution as well as the value and benefits the team members provide the prime is also an objective of this RFSB. Submittals will inform RCCTO as to the technical and competitive landscape that exists for capabilities, concepts, prototypes, or solutions that can meet or exceed the desired software requirements and program objectives.

General Capability Description.

The overall objective of HMIF is to increase the lethality and survivability of our maneuver formations by integrating the best of breed robotic and autonomous systems (RAS) hardware and software to enhance warfighting functions. To that end, this RFSB is seeking industry partners that can self-organize into effective teams to design, develop, and integrate HMIF RAS Software that enables seamless operation of air and ground robotic systems and mission command interoperability to enhance lethality and survivability of Army formations. The RFSB further supports development and integration of HMIF Robotics RAS Control and Autonomy Software within the U.S. Army's broader HMIF Architecture to integrate RAS in operational Formations to fight in contested operational environments. The proposed solution should provide modular, scalable, and standards-based HMIF software architectures, capable of safely controlling ground and air unmanned systems during mobility and lethality teleoperation, and autonomous functions while interoperating and integrating with emerging Army mission command frameworks like the Next Generation Command and Control (NGC2) initiative.

Development Timeline and Technology Readiness Level (TRL): The Army is seeking HMIF RAS Software prototype solutions, requiring a limited amount of development that can be integrated, tested, validated, and demonstrated as an operational software prototype at TRL 7+ (i.e., system prototype demonstration in an operational environment). As part of the RFSB, respondents are requested to estimate a period of performance (POP) and project schedule for their proposed approach to reach or exceed TRL 7 or above.

Solution Brief Guidelines. Eligible interested parties can respond to this RFSB with a technical solution write-up. Solution Briefs shall not exceed six (6) one-sided pages plus a cover sheet (maximum of 6-pages) with company information. Solution Briefs shall be single-spaced, 12-point type with at least one-inch margins on 8 1/2" x 11" page size. Pictures or graphic representations do not count against the 6-page limit. Total file size shall not exceed a 10 MB e-mail limit for all items associated with this RFSB response. Solution Briefs must specifically address the requested information. Oral communications initiated by the respondent are not permissible during this phase. If company proprietary information is included within a response, the proprietary portions of the response must be so marked. Solution Briefs should outline both capabilities and technical software solutions that address and meet or exceed the stated requirements in the attached SOO.

- The 6-page solution brief, in contractor format, may focus on describing details of the proposed software solution, including how it could address or improve the technical

requirements and objectives, and potential impacts and benefits to the HMIF program. While there is no guarantee of a future request for prototype proposal, solution briefs should state whether the contractor would participate in future pitches or prototype proposals for the concepts outlined in their technical write up. RCCTO is seeking solutions from industry to assist in meeting the HMIF Control and Autonomy Software requirements.

- The Solutions Brief should be outlined as follows:
 - Cover Page to include Company name and address, and CAGE Code; whether it is a traditional or non-tradition defense contractor (defined below); the Technical and Business Points of Contact (POC) to include: Telephone Numbers and Email Addresses; headquarters location; size designation of company (i.e., large or small business concern), annual revenues, and number of employees; if the Company's proposed approach has been proposed to or previously funded by any federal agency, to include SBIR/STTR, please identify each specific agency and all applicable contract/agreement numbers; and a brief abstract.
 - Pages 1-2: Technical Solution/Approach. The Technical Solution/Approach should address the Government's needs as outlined in the SOO and how the solution can meet or exceed stated objectives; provide supporting evidence and demonstration of software component and RAS Software Integration Technology Readiness Level exit of TRL 7+ or higher; align with HMIF Architecture and Defined Standards and Interfaces; and discuss the security approach and maturity of cyber Risk Management Framework (RMF) procedures to support ATO process schedule.
 - Page 3: Teaming Arrangement. The Teaming Arrangement discussion should briefly discuss the team composition for executing the objectives of the SOO and outline which team members will perform which development efforts. A brief discussion on the team's experience in uncrewed vehicle control and DevSecOps is preferred.
 - Page 4: Schedule. The Schedule should outline the development approach and schedule for logical sequencing and achievability; demonstration of a tested, demonstrated, and validated solution within a nominal 4 (O) to 6 (T) month period with a goal of integrating, testing and validating with HMIF platforms, systems, networks, and payloads in that timeframe, or shortly thereafter; and consideration of reuse of existing Government developed control, autonomy and assured control/safety software modules.
 - Page 5: Data Rights Assertions if applicable.
 - Page 6: Rough Order of Magnitude (ROM). The ROM should provide the Government with a rough order of magnitude necessary to complete the solution.

Solution Briefs shall be clearly marked “Company Name_RFSB_HMIF-RAS SW Solution.” Only UNCLASSIFIED submissions will be accepted. Any submissions with classified information will not be reviewed and will be deleted. There will be no mailed submissions and telephone inquiries will not be accepted. Review of solution briefs does not promise nor guarantee that a company will receive a contract, and the Government reserves the right, at any point, to cancel the RFSB. Please be advised, only an Agreements Officer has the authority to enter into, or modify, a binding agreement on behalf of the Government.

Other Transaction Agreement. RCCTO intends to exercise the authority at 10 U.S.C. § 4022 for an Other Transaction Agreement prototype project. This RFSB and any subsequent activities are intended to satisfy competition requirements of the statute. This phased competition, executed under the authorities of 10 U.S.C. § 4022, will consist of the following three (3) phases in the review and selection process as follows:

- Phase 1, A Request for Solutions Brief: six (6) pages plus cover page of a technical discussion of a software development solution, any teaming arrangements, a rough project development schedule, data rights assertions (if applicable), and rough order of magnitude (ROM) price based on the attached SOO and evaluation criteria.
- Phase 2, Pitch Session: Companies whose Solution Briefs are evaluated to be of merit may be invited to provide a pitch either in-person or virtual to demonstrate and provide further details on their proposed concept/technology/solution and the technical and business viability of the solution to RCCTO’s subject matter expert panel, consisting of representation across the RCCTO and Army/DoD Community of Interest. The Government reserves the right to limit the number of Companies invited to pitch. As such, a Company’s solution may be evaluated to be of merit but not invited to pitch.
- Phase 3, Proposal: Those Companies whose Solution Brief and Pitch are evaluated to be of merit and are selected for potential award of a Prototype OTA may, if funding is available, be invited to submit a full written proposal. This phase will include development of a written proposal and negotiation of terms and conditions governing the prototype project to include the collaborative development of a Statement of Work.

RFSB submissions will be due at **12:00 PM/1200 ET on Friday, 16 May 2025**, via the following RCCTO email address:

- Jolynda Ivy at jolynda.v.ivy.civ@army.mil
- Tessa Jones at tessa.a.jones.civ@army.mil
- Simone Brightmon at simone.l.brightmon.civ@army.mil

Key Milestones are listed in the table below.

RFSB Release Date	25 April 2025
RFSB Questions	1 May 2025
RFSB Responses Due	16 May 2025
Invitation to Pitch Session	27 May 2025
Pitch Session	05 June 2025
RFPP Release Date	TBD
OTA Award	TBD

Phase 1, Request for Solutions Brief Basis of Evaluation:

All solution briefs correctly submitted in response to this announcement will be evaluated against the stated criteria below:

1. Technical merit, feasibility and relevance of the Solution Brief in addressing the SOO.
2. Company's approach and/or underlying technology to the Government application, including the planned development schedule.
3. The ROM is affordable/realistic.

Solution Briefs will be evaluated on the basis of merit of the proposed solution/concept in addressing the SOO, not against other Solution Briefs submitted in response to this announcement. However, following initial evaluation, RCCTO may select Companies who submit solution briefs with particular merit or benefit to provide an oral presentation/pitch.

Phase 2, Oral Presentation/Pitch Evaluation Criteria: Individual pitches will be evaluated against the following factors:

1. Relevance of the pitch in addressing the SOO.
2. The solution has technical merit, adequately addresses Government needs and is feasible;
3. The ROM;
4. The notional schedule is realistic and acceptable; and
5. The potential impact of data rights assertions.

Individual pitches will be evaluated on the basis of merit of the proposed concept in addressing the SOO and the factors above, not against any other pitches held under this announcement. Pitch evaluation will consider the full content of the actual Oral Presentation (including answers or inability to answer any questions posed) and the content of the slides presented. Pitch evaluation will also consider the previously submitted solution brief; thus, inconsistencies between the solution brief and presentation, or between the digital copies of the slides and the oral presentation itself, may be viewed unfavorably. It is anticipated that pitches will be conducted virtually with an up to one hour presentation and up to 30-minute question/answer session.

Questions. The Government will consider and answer relevant questions to this RFSB. Submit questions to the POCs identified above NLT 1200 ET on 1 May 2025. Depending on the nature of the question, the Government may respond directly to the inquiry or post questions and answers as an amendment to this announcement.

Non-Government Advisors. All government or contractor personnel that may review RFSB responses have executed Non-Disclosure Agreements (NDAs) for project information related to this initiative.

Eligibility Requirements. Only U.S.-based companies or organizations are considered eligible to participate in this RFSB. Submissions from entities or organizations that do not have a primary place of business in the U.S. will not be considered.

Each eligible entity:

- U.S.-based firms shall be incorporated in, and maintain, a primary place of business in the U.S.
- May not be a U.S. Federal government entity or employ a U.S. Federal employee acting within the scope of their employment.
- Is required to register in System for Award Management (SAM) and have, or be able to obtain, an active CAGE code, DUNS number, and not be on the exclusion lists (debarred, suspended, or ineligible).

RCCTO's RFSB is voluntary and open to all entities that meet the eligibility requirements. Submittals to the RFSB must be received **by 12:00 PM ET on 16 May 2025**. Late submissions will not be accepted, and any other failure to comply with RFSB submission instructions (content, approach, etc.) may not be evaluated by the Government.

Intellectual Property. The Army is a strong proponent of deliberate intellectual property (IP) rights and management by the private sector and the Department of Defense.

For the RFSB:

- Nothing in this RFSB shall diminish the Government's rights in patents, technical data, technical information, computer software, computer databases, and computer software documentation that the Government had prior to this effort, or is entitled to, under any other Government Agreement or contract, or is otherwise entitled to under law; and
- The Federal Government may negotiate licenses for the use of IP developed by a registered participant in the future.

Furthermore, the Government understands that information provided in response to this solicitation is presented in confidence and may contain trade secret or commercial or financial information, and it agrees to protect such information from unauthorized disclosure to the maximum extent permitted or required by Law, to include:

- a. 18 USC 1905 (Trade Secrets Act),
- b. 18 USC 1831 et seq. (Economic Espionage Act),
- c. 5 USC 552(b)(4) (Freedom of Information Act),
- d. Executive Order 12600 (Pre-disclosure Notification Procedures for Confidential Commercial Information), and

e. Any other statute, regulation, or requirement applicable to Government employees.

Notes & Definitions.

Non-traditional defense contractor is defined in 10 U.S.C. § 3014 as an entity that is not currently performing and has not performed, for at least the one-year period preceding the solicitation of sources by the Department of Defense for the procurement or transaction, any contract or subcontract for the Department of Defense that is subject to full coverage under the cost accounting standards prescribed pursuant to section 1502 of title 41 and the regulations implementing such section.

Prototype projects are defined as submissions that may improve Army platforms, Army weapons systems, or Army support systems. A prototype project can generally be described as a preliminary pilot, test, evaluation, demonstration, or agile development activity used to evaluate the technical or manufacturing feasibility or military utility of a particular technology, process, concept, end item, effect, or other discrete feature. Prototype projects may include systems, subsystems, components, materials, methodology, technology, or processes. By way of illustration, a prototype project may involve: a proof of concept; a pilot; a novel application of commercial technologies for defense purposes; a creation, design, development, demonstration of technical or operational utility; or combinations of the foregoing, related to a prototype.

DISCLAIMER:

All information contained in this RFSB is preliminary as well as subject to modification and is in no way binding on the Government. This notice and the information presented herein will not obligate the Government in any manner. The Government does not intend to pay for information received in response to this RFSB. Responders to this announcement are solely responsible for all expenses associated with responding to this RFSB. This RFSB will be the basis for collecting information on capabilities available. Please be advised that all submissions become Government property and will not be returned nor will receipt be confirmed.